

IVVAVIK NATL PARK, YT, Canada

WMO: 719780

Lat: 69.164N

Lon: 140.150W

Elev: 244

StdP: 98.43

Time Zone: -8.00 (NAP)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-39.1	-37.2	-43.8	0.1	-39.0	-41.8	0.1	-37.0	13.6	-15.8	12.3	-16.4	1.8	200	0.885

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	24.0	15.1	21.9	14.0	20.0	13.2	15.8	22.1	14.7	20.6	13.7	19.1	4.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.9	9.5	18.1	11.8	8.9	17.0	10.7	8.3	16.1	45.1	22.1	42.0	20.6	39.1	19.2	19.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	10.9	9.2	7.7	DB	-41.4	27.7	2.2	1.5	-42.9	28.8	-44.2	29.7	-45.5	30.6	-47.1	31.7
(5)				WB	-41.5	17.5	2.2	1.2	-43.0	18.3	-44.3	19.0	-45.6	19.7	-47.2	20.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-7.9	-24.0	-22.8	-21.9	-11.2	0.8	9.9	12.4	8.7	2.7	-9.2	-18.8	-22.8
	DBStd	14.99	8.09	7.95	6.88	7.19	6.31	4.58	4.01	4.24	4.81	6.04	6.57	7.03
	HDD10.0	6741	1053	918	989	637	288	57	22	77	223	596	864	1017
	HDD18.3	9592	1311	1152	1247	887	542	254	186	301	470	854	1114	1275
	CDD10.0	195	0	0	0	0	4	54	98	36	3	0	0	0
	CDD18.3	4	0	0	0	0	0	1	3	1	0	0	0	0
	CDH23.3	83	0	0	0	0	4	23	44	13	0	0	0	0
	CDH26.7	8	0	0	0	0	0	2	5	1	0	0	0	0
Wind	WSAvg	3.3	3.4	3.6	3.3	3.4	3.8	3.6	3.2	3.0	2.9	2.8	3.2	3.0
Precipitation	PrecAvg	249	12	9	9	7	13	30	46	46	28	22	14	12
	PrecMax	313	33	22	23	21	32	62	89	92	55	38	29	29
	PrecMin	190	3	2	1	1	3	8	19	12	14	9	2	3
	PrecStd	32	7	4	5	4	6	14	13	16	9	8	7	6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.4	-2.3	-0.1	9.3	22.3	26.2	27.0	25.1	17.5	6.8	-0.2	-2.8
		MCWB	-2.7	-4.3	-2.4	4.3	12.2	15.0	16.8	16.4	10.7	3.6	-2.4	-4.5
	2%	DB	-5.2	-4.9	-4.9	5.9	18.0	23.2	24.4	21.2	14.9	3.0	-3.9	-7.1
		MCWB	-6.8	-6.4	-6.7	2.2	10.1	13.8	15.5	14.1	9.2	0.8	-5.5	-8.3
	5%	DB	-8.5	-8.8	-8.7	2.8	14.5	21.2	22.4	18.8	12.5	0.9	-7.1	-9.9
		MCWB	-9.5	-10.1	-10.1	-0.2	8.1	13.1	14.7	12.8	8.2	-0.7	-8.2	-10.8
	10%	DB	-12.1	-11.4	-11.5	0.0	11.3	19.1	20.5	16.4	10.2	-1.1	-9.7	-13.1
		MCWB	-13.0	-12.4	-12.6	-2.2	6.2	12.0	14.1	11.6	6.9	-2.4	-10.6	-13.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.7	-4.1	-2.2	4.5	12.7	16.2	17.7	17.0	11.2	4.1	-2.4	-4.4
		MCDB	-0.6	-2.5	-0.2	8.7	21.8	23.7	25.3	24.0	16.3	6.2	-0.5	-2.8
	2%	WB	-6.7	-6.4	-6.7	2.1	10.4	14.7	16.2	14.6	9.7	1.1	-5.4	-8.3
		MCDB	-5.5	-4.9	-5.1	5.7	17.3	21.9	22.5	19.9	14.1	2.7	-4.0	-7.1
	5%	WB	-9.6	-10.0	-10.1	-0.1	8.4	13.5	15.3	13.2	8.4	-0.6	-8.2	-10.8
		MCDB	-8.4	-8.8	-8.8	2.6	14.4	20.1	21.2	17.8	12.2	0.8	-7.2	-9.8
	10%	WB	-13.0	-12.4	-12.6	-2.2	6.3	12.4	14.4	12.0	7.0	-2.3	-10.6	-13.8
		MCDB	-12.1	-11.4	-11.5	-0.1	11.1	18.5	19.8	16.1	9.9	-1.1	-9.7	-13.0
Mean Daily Temperature Range	5% DB	MDBR	7.4	7.6	9.8	11.2	10.6	12.0	11.0	9.7	8.5	7.2	6.9	7.3
		MCDBR	10.6	9.8	10.8	11.8	16.7	15.0	14.6	13.9	12.5	7.3	8.5	9.7
		MCWBR	9.4	8.8	9.4	8.5	9.6	7.3	7.2	7.5	5.8	7.6	9.0	
	5% WB	MCDBR	10.7	10.2	10.6	11.3	16.6	14.0	13.6	12.6	11.9	6.9	8.3	9.4
		MCWBR	9.5	9.2	9.2	8.2	9.7	7.1	7.1	7.0	7.2	5.6	7.6	8.8
Clear-Sky Solar Irradiance	taub	0.167	0.206	0.248	0.306	0.336	0.353	0.374	0.339	0.292	0.216	0.153	N/A	
	taud	1.896	2.114	2.213	2.113	2.172	2.351	2.359	2.466	2.478	2.178	1.918	N/A	
	Ebn at Noon	52	591	791	825	842	841	805	800	749	557	50	0	
	Edh at Noon	4	40	72	110	118	103	99	79	58	35	3	0	
All-Sky Solar Radiation	RadAvg	0.02	0.43	2.08	4.41	5.95	6.13	4.92	3.29	1.84	0.60	0.07	0.00	
	RadStd	0.00	0.04	0.11	0.21	0.39	0.56	0.37	0.29	0.19	0.08	0.01	0.00	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.83	N/A	N/A	N/A	N/A	N/A	-286	-301	N/A	N/A
(47)	Regional (7 neighbors)	+0.87	+0.92	N/A	N/A	N/A	N/A	-315	-324	N/A	N/A

Nomenclature: See separate page